

Vector Semantic and Embeddings

Introduction

Nets are for fish; 言者所以在意，得意而忘言

Once you get the fish, you can forget the net.

Words are for meaning; 荃者所以在鱼，得鱼而忘荃

Once you get the meaning, you can forget the words

庄子 (Zhuangzi) Chapter 26

Lexical Semantic

Distributional Hypothesis

Definition – words that occur in *similar contexts* tend to have similar meanings. The link between how words are distributed and the similarity in what they mean is called the **distributional hypothesis**.

- Formulated by linguistics like Joos (1950), Harris (1954), and Firth (1957)

Examples:

Eye-doctor & Oculist -> near words are **eye** or **examined**

Lexical Semantic – Lemmas and Senses(1/6)

Mouse (N)

1. any of numerous small rodents...
2. a hand-operated device that controls a cursor...

Mouse is lemma for: Mice and Mouse

Word sense: each of the aspects of the meaning of a word is called **word sense**

Lexical Semantic – Synonym (2/6)

Synonym – when one sense of a word is identical or near identical to one sense in another word then these two words are synonyms.

- *couch/sofa*
- *vomit/throw up*
- *filbert/hazelnut*
- *car/automobile*
- *H₂O/water*

Lexical Semantic – Word Similarity (3/6)

Word Similarity – words may not have so many synonyms but certainly have similar words.

- *Cat and Dog are similar words*

Manual Annotation of Similarity (SimLex-999 dataset, Hill et. Al 2015)

vanish	disappear	9.8
belief	impression	5.95
muscle	bone	3.65
modest	flexible	0.98
hole	agreement	0.3

Lexical Semantic – Word Relatedness (4/6)

Word relatedness or association in psychology – two words are co-participating in an event or are related eventively. A common way that words can be related is through the same ***semantic field***, which is a set of words that cover a particular semantic domain and bear structured relations with each other.

- Co-participation: *cup* and *coffee*
- Related Eventively: *Scalpel* and *Surgeon*
- Semantic Field:
 - field of **hospitals**: *surgeon, scalpel, nurse, anesthetic, hospital*
 - Field of **restaurants**: *waiter, menu, plate, food, chef*

Lexical Semantic – Semantic Frames and Roles (5/6)

Semantic Frames – closely related to semantic fields is semantic frames. A semantic frame is a set of words that denote **perspectives** or **participants** in a particular type of event.

- *buy* (the event from the perspective of the buyer)
- *sell* (from the perspective of the seller)
- *pay* (focusing on the monetary aspect)

Paraphrasing Task:

Sam bought the book from Ling (Paraphrase to) Ling sold the book to Sam

Lexical Semantic – Connotation(6/6)

Connotation - Words have *affective meanings* or *connotations*. The word connotation has different meanings in different fields, but here we use it to mean the aspects of a word's meaning that are related to a writer or reader's emotions, sentiments, opinions, or evaluations.

- Pos-connotation: *wonderful*; Neg-connotation: *dreary*
 - A mix of Pos-conn and Neg-conn of words with the same meaning
 - *copy, replica, reproduction*
- Pos-evaluation: *great, love*; Neg-evaluation: *hate, terrible*
- Words are different in 3 dimensions of *affect* (Osgood 1957):
 - **valence**: the pleasantness of the stimulus
 - **arousal**: the intensity of emotion provoked by the stimulus
 - **dominance**: the degree of control exerted by the stimulus



Vector Semantic

Ongchoi is delicious sauteed with garlic.

Ongchoi is superb over rice.

... ongchoi leaves with salty sauces ...

And suppose that you had seen many of these context words in other contexts:

... spinach sauteed with garlic over rice ...

... chard stems and leaves are delicious ...

... collard greens and other salty leafy greens

Vector Semantic



T-SNE plot of some word embedding in 2 dimensions space

Words and Vectors

Vectors or the Distributional model of word meaning is based on a co-occurrence matrix, a way to represent how often two words co-occur. Two popular matrices:

- Term-document co-occurrence matrix
- Term-term co-occurrence matrix

Words and Vectors

	As You Like It	Twelfth Night	Julius Caesar	Henry V
battle	1	0	7	13
good	114	80	62	89
fool	36	58	1	4
wit	20	15	2	3

4 documents from Shakespeare's plays and 4 words

- **Vector** – a list or array of numbers
- Julius Caesar -> [7, 62, 1, 2]

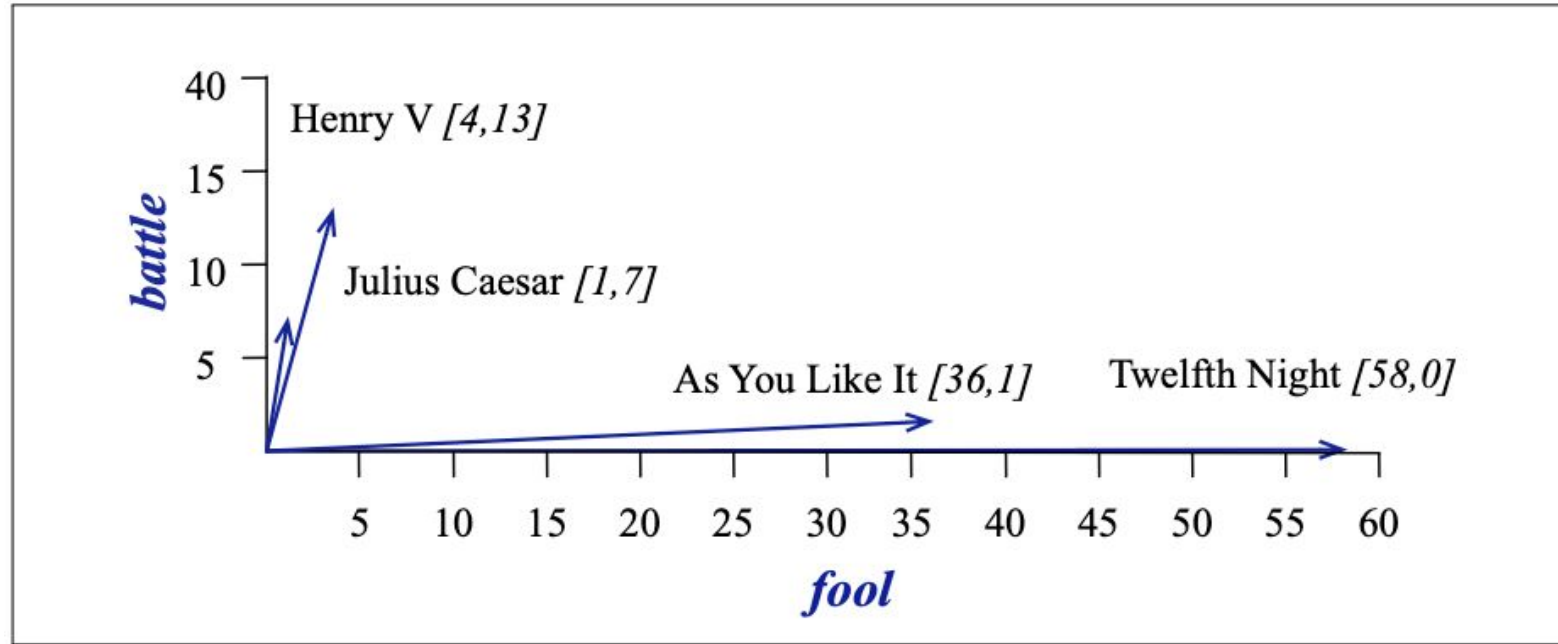
Words and Vectors

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4 documents from Shakespeare's plays and 4 words

- **Vector** – a list or array of numbers
- Julius Caesar -> [7, 62, 1, 2]
- Column Vector – the red highlight around Julius Caesar

Words and Vectors



Visualization of 4 of Shakespeare's plays documents showing only 2-dimensions for each document

Words as vectors: document dimensions

	As You Like It	Twelfth Night	Julius Caesar	Henry V
battle	1	0	7	13
good	114	80	62	89
fool	36	58	1	4
wit	20	15	2	3

The term-document matrix for four words in four Shakespeare plays. The red boxes show that each word is represented as a row vector of length four.

Battle -> [1, 0, 7, 13]

Words as vectors: word dimensions (1/3)

Context window size ± 4 words around each word

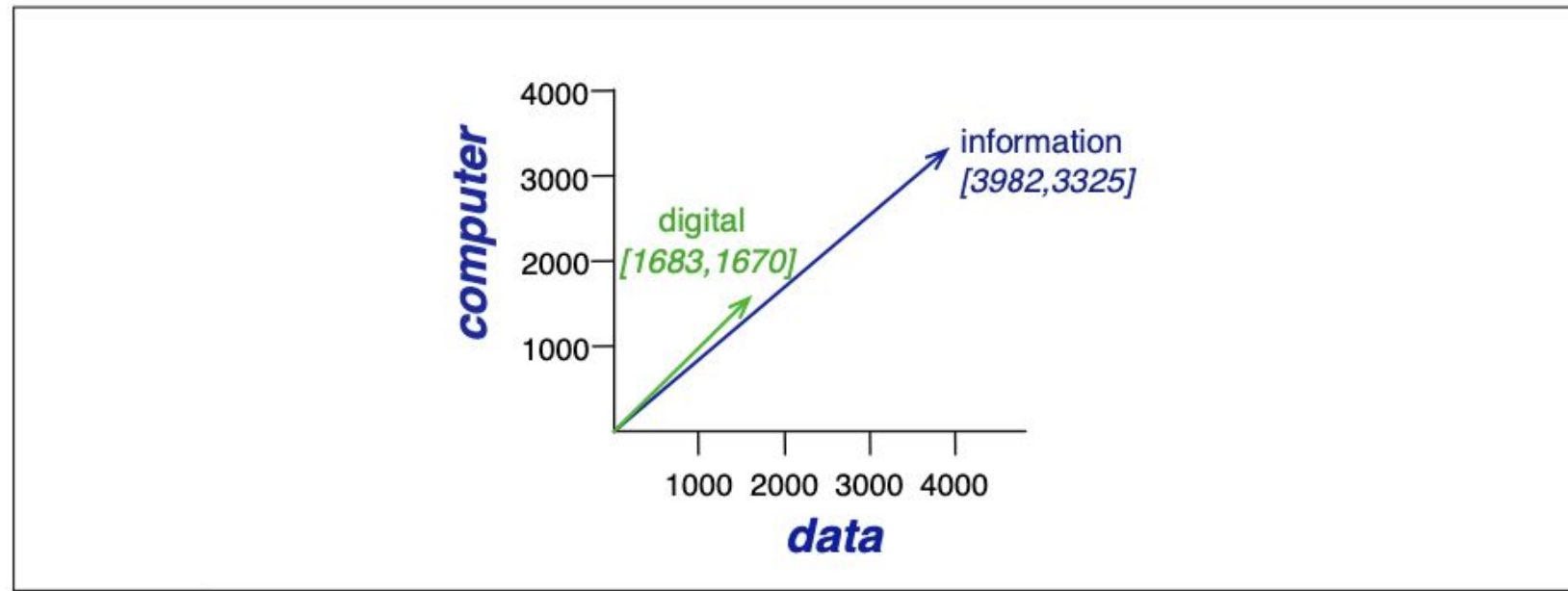
*is traditionally followed by **cherry** pie, traditional dessert
often mixed, such as **strawberry** rhubarb pie, Apple pie
computer peripherals and personal **digital** assistants, These devices usually
a computer. This includes **information** available on the internet*

Words as vectors: word dimensions (2/3)

	aardvark	...	computer	data	result	pie	sugar	...
cherry	0	...	2	8	9	442	25	...
strawberry	0	...	0	0	1	60	19	...
digital	0	...	1670	1683	85	5	4	...
information	0	...	3325	3982	378	5	13	...

Co-occurrence vectors for 4 words in Wikipedia Corpus, showing six of the dimensions.

Words as vectors: word dimensions (3/3)



A spatial visualization of word vectors for *digital* and *information*, showing just two of the dimensions, corresponding to the words *data* and *computer*.

Cosine for Measuring Similarity (1/2)

$$\text{dot product}(\mathbf{v}, \mathbf{w}) = \mathbf{v} \cdot \mathbf{w} = \sum_{i=1}^N v_i w_i = v_1 w_1 + v_2 w_2 + \dots + v_N w_N$$

Vector Length:

$$|\mathbf{v}| = \sqrt{\sum_{i=1}^N v_i^2}$$

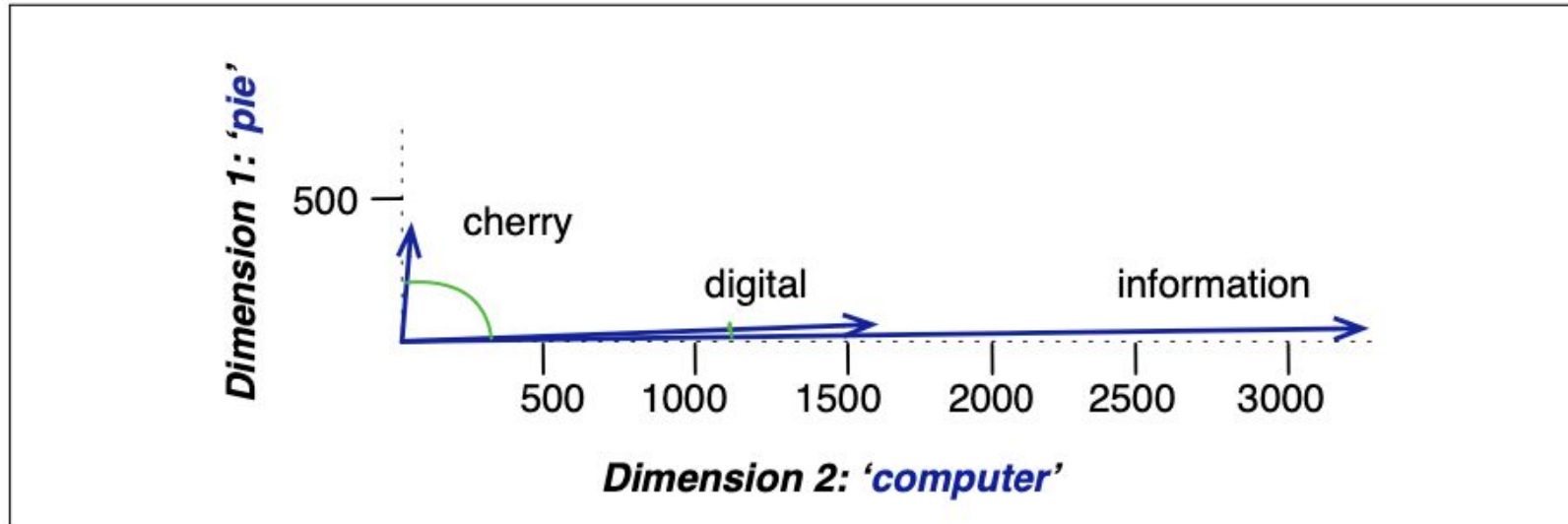
Normalized dot product:

$$\begin{aligned} \mathbf{a} \cdot \mathbf{b} &= |\mathbf{a}| |\mathbf{b}| \cos \theta \\ \frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{a}| |\mathbf{b}|} &= \cos \theta \end{aligned}$$

Cosine Similarity:

$$\text{cosine}(\mathbf{v}, \mathbf{w}) = \frac{\mathbf{v} \cdot \mathbf{w}}{|\mathbf{v}| |\mathbf{w}|} = \frac{\sum_{i=1}^N v_i w_i}{\sqrt{\sum_{i=1}^N v_i^2} \sqrt{\sum_{i=1}^N w_i^2}}$$

Cosine for Measuring Similarity (2/2)



A (rough) graphical demonstration of cosine similarity, showing vectors for three words (*cherry*, *digital*, and *information*) in the two-dimensional space defined by counts of the words *computer* and *pie* nearby.

	pie	data	computer
cherry	442	8	2
digital	5	1683	1670
information	5	3982	3325

$$\cos(\text{cherry}, \text{information}) = \frac{442 * 5 + 8 * 3982 + 2 * 3325}{\sqrt{442^2 + 8^2 + 2^2} \sqrt{5^2 + 3982^2 + 3325^2}} = .018$$

$$\cos(\text{digital}, \text{information}) = \frac{5 * 5 + 1683 * 3982 + 1670 * 3325}{\sqrt{5^2 + 1683^2 + 1670^2} \sqrt{5^2 + 3982^2 + 3325^2}} = .996$$

TF-IDF: Weighing terms in the vector

Term frequency raw
count:

$$\text{tf}_{t,d} = \text{count}(t,d)$$

Adding the *log* to
normalize high
frequency and zero
count:

$$\text{tf}_{t,d} = \begin{cases} 1 + \log_{10} \text{count}(t,d) & \text{if } \text{count}(t,d) > 0 \\ 0 & \text{otherwise} \end{cases}$$

Document Frequency:

$$\text{idf}_t = \log_{10} \left(\frac{N}{\text{df}_t} \right)$$

N is the # of the total documents

TF-IDF:

$$w_{t,d} = \text{tf}_{t,d} \times \text{idf}_t$$

TF-IDF: Weighing terms in the vector

	As You Like It	Twelfth Night	Julius Caesar	Henry V
battle	0.246	0	0.454	0.520
good	0	0	0	0
fool	0.030	0.033	0.0012	0.0019
wit	0.085	0.081	0.048	0.054

A tf-idf weighted term-document matrix for four words in four Shakespeare plays, using the counts

Dense Vector Representation (word2vec)

Model for Dense Vector Representation

- **Documents** = {"This is an example", "This is another example"}
Vocabulary = {this, is, an, another, example}
- For a window size of 1, the surrounding words around "is" in the first document are "this" and "an"
- We can build a binary classifier for pairs (*input word*, *output word*) such that
 - The pairs (**is**, **this**) and (**is**, **an**) are classified as **true** (neighboring words)
 - The pairs (**is**, **example**) and (**is**, **another**) are classified as **false**
- Let $\mathbf{w}_i = [w_{i1}, w_{i2}, \dots, w_{iD}]$ be the vector representation of word i in D -dimension



- Neural network can be used to train $\mathbf{w}_i$

Word Vectors: Learning with Neural Networks

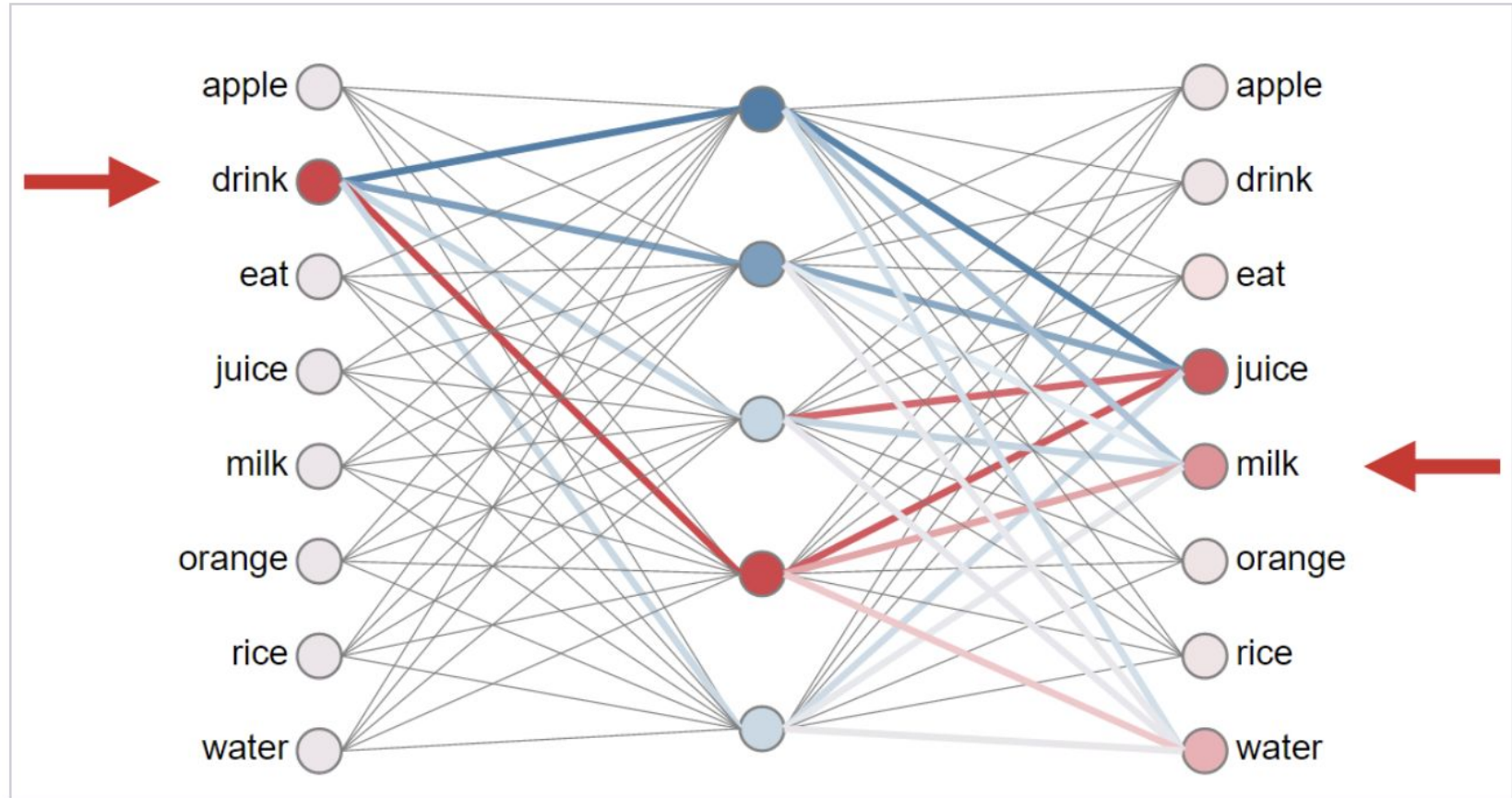
Neural
Network-based
Word Embedding
Models

Google™

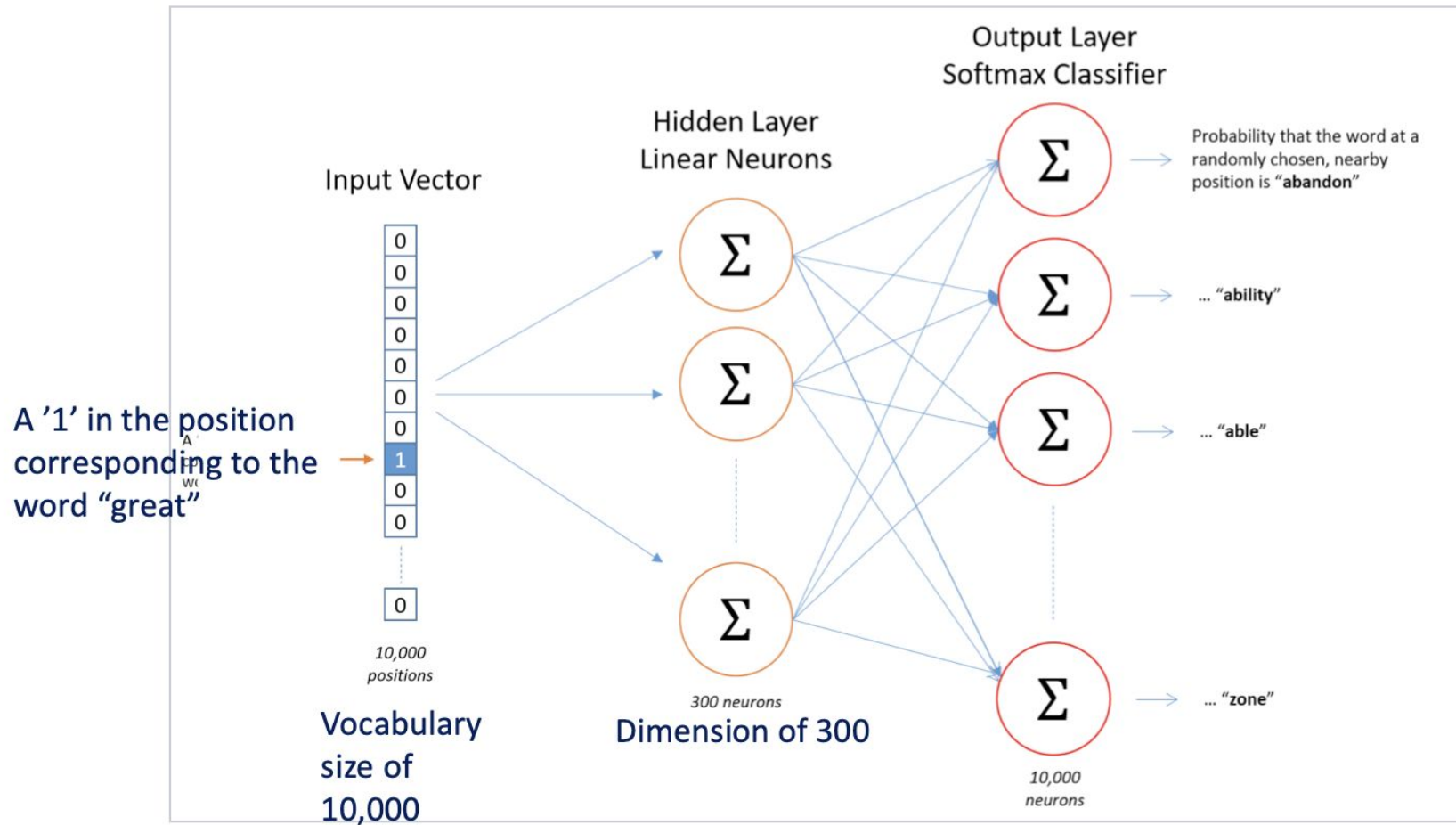
Word2vec

facebook

FastText



Word2Vec Neural Network Architecture



Word2Vec Learning Models

Continuous Bag of Words (CBOW):

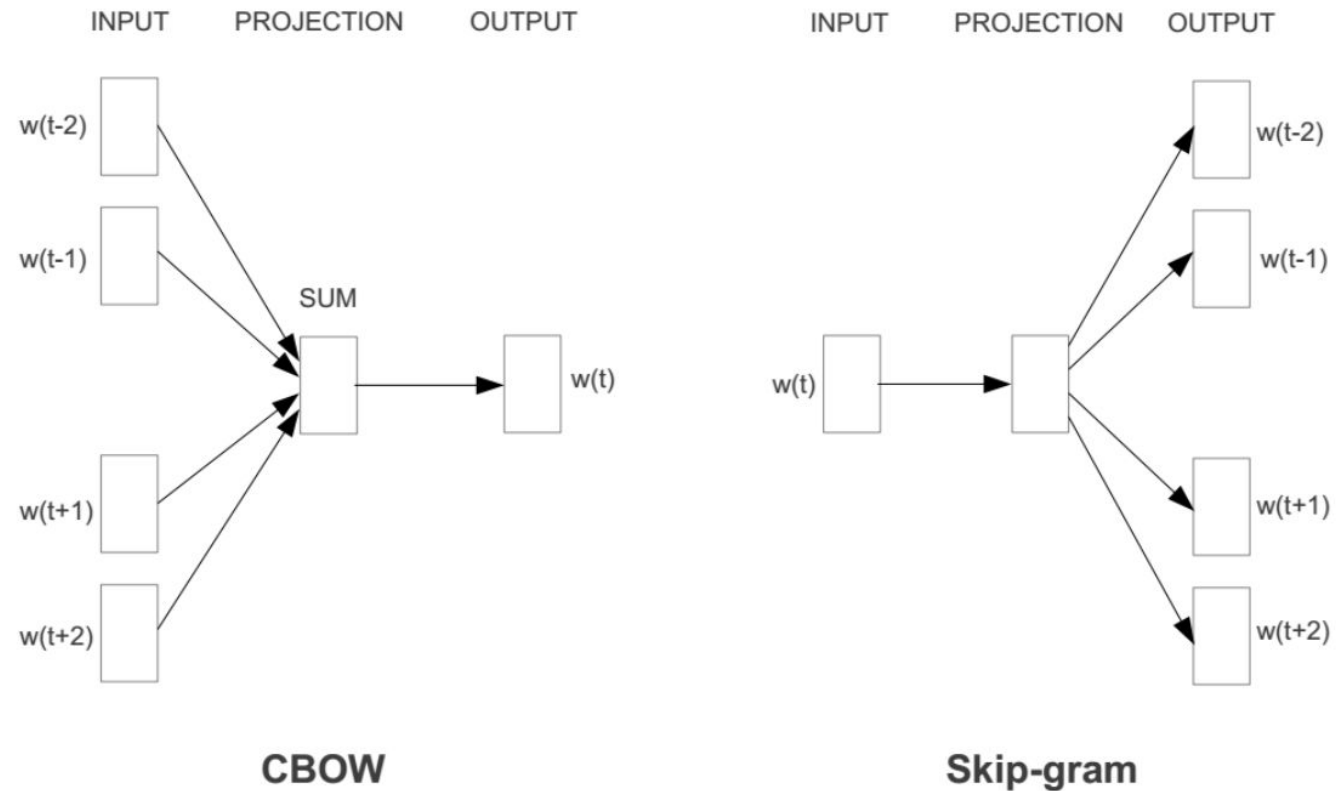
- Given the context predict the word:
 - $w_{i-2}, w_{i-1}, w_i, w_{i+1}, w_{i+2}$
- Example: **The cat ate** _____.
 - Fill in the blank, e.g. “food”.
- Faster to train
- Works well for large amount of training data

Continuous Skip-Gram:

- Given the word predict the context:
 - $w_{i-2}, w_{i-1}, w_i, w_{i+1}, w_{i+2}$
- Ex: ____ food.
 - Fill in the blank, e.g. “The cat ate”
- Slower to train
- Works better for infrequent words

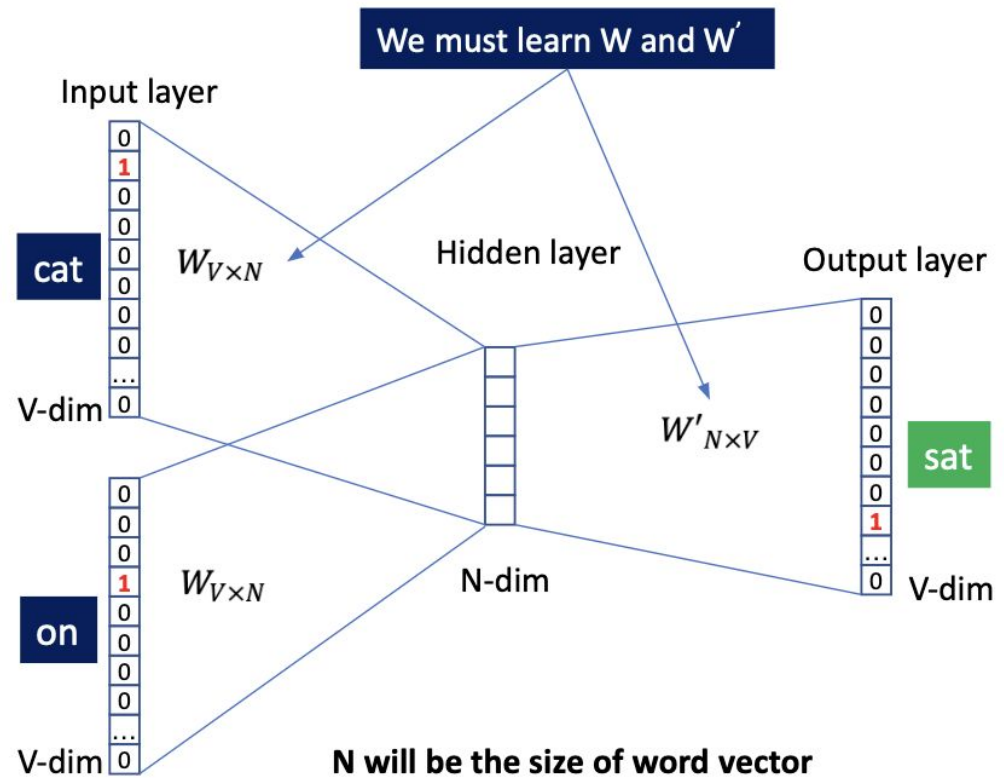
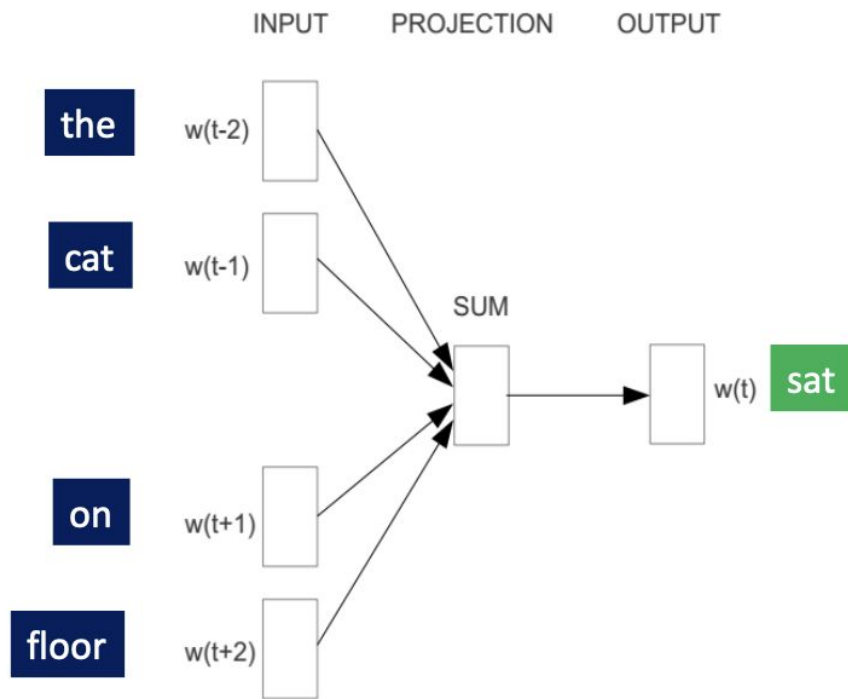
Word2vec

- Represent the meaning of the word by its context

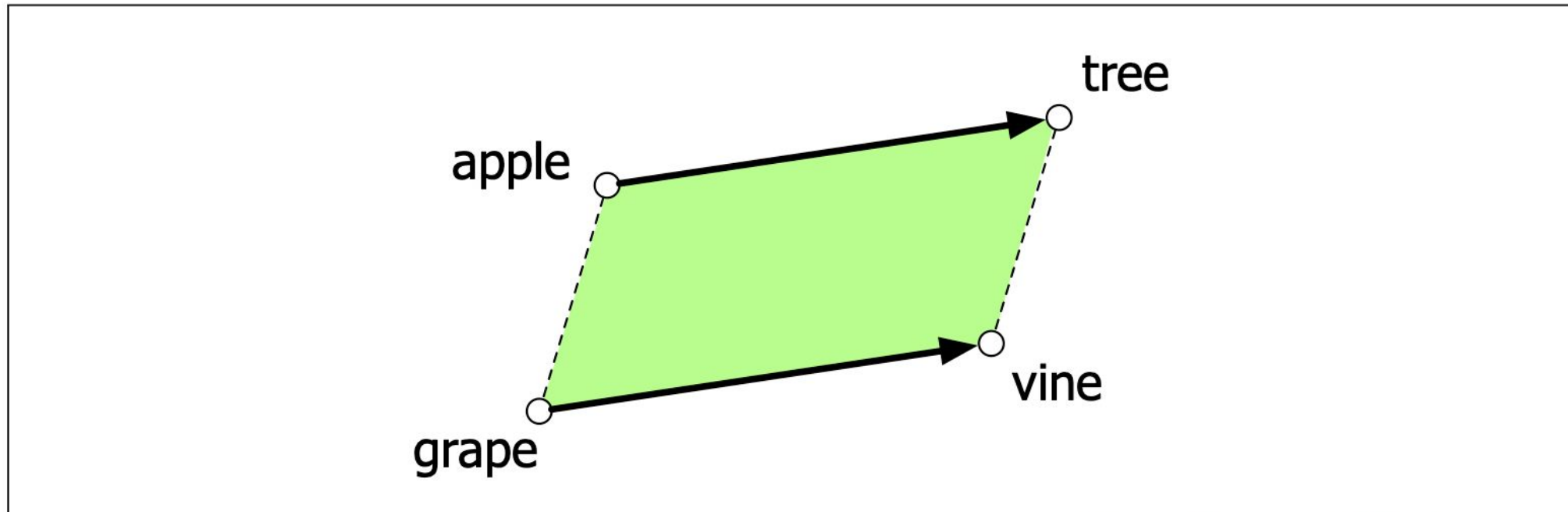


Word2Vec: CBOW Example

- "the cat ____ on floor"

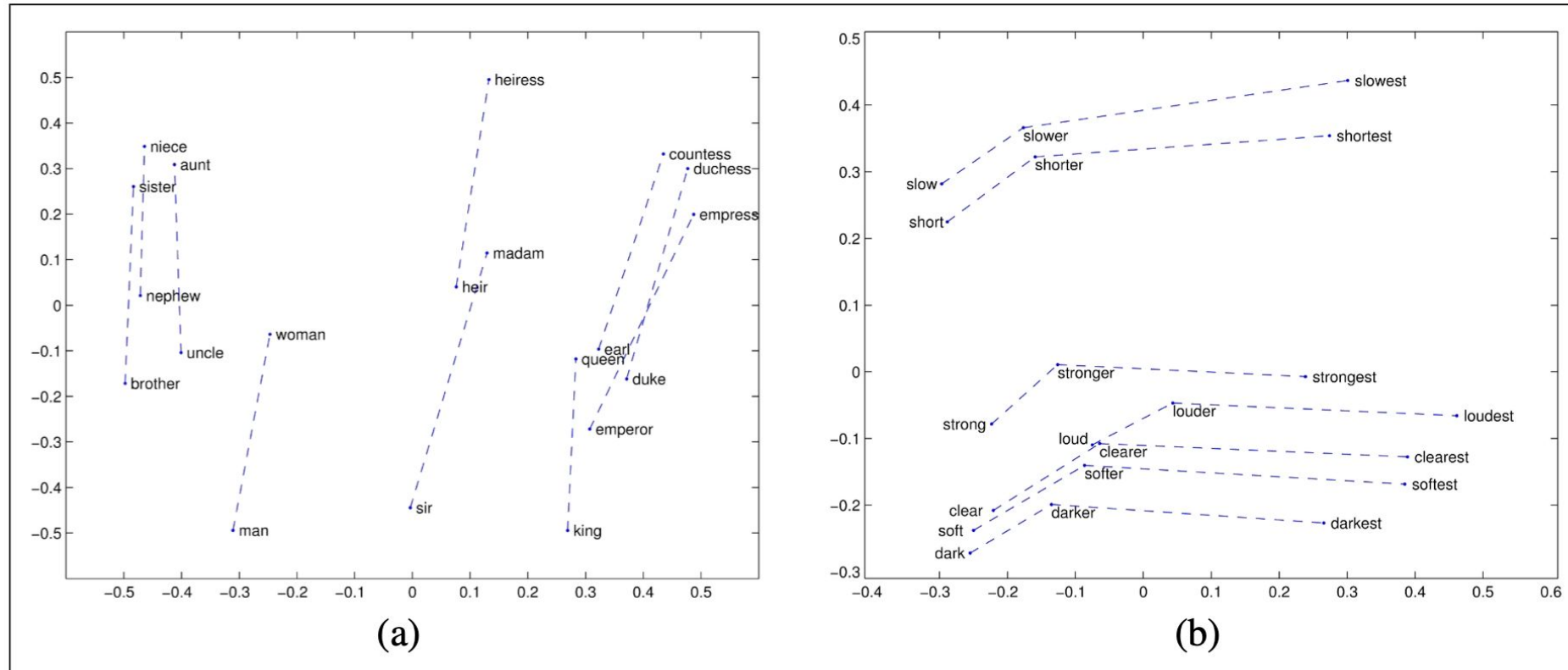


Semantic Properties of Embeddings



The parallelogram model for analogy problems ([Rumelhart and Abrahamson, 1973](#)): the location of $\vec{\text{vine}}$ can be found by subtracting $\vec{\text{apple}}$ from $\vec{\text{tree}}$ and adding $\vec{\text{grape}}$.

Semantic Properties of Embeddings



Relational properties of the GloVe vector space, shown by projecting vectors onto two dimensions. (a) $\vec{\text{king}} - \vec{\text{man}} + \vec{\text{woman}}$ is close to $\vec{\text{queen}}$. (b) offsets seem to capture comparative and superlative morphology (Pennington et al., 2014).